Azure Cloud Security Lab

Hands-On Identity, Security & Monitoring Project

Author	Platform	Date	Labs Completed
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Technologies Used

Identity & Access	Security Monitoring	Infrastructure
Microsoft Entra ID	Microsoft Sentinel (SIEM)	Azure Virtual Network
Azure Key Vault	Microsoft Defender for Cloud	Network Security Groups
RBAC & IAM	KQL Detection Queries	Azure Monitor & Alerts

Project Overview

This project demonstrates hands-on implementation of cloud security controls across Microsoft Azure. Each lab covers a critical security domain relevant to enterprise environments including identity and access management, secret management, threat detection, security posture management, network security, and automated alerting. All labs were performed live in a real Azure environment using a personal subscription.

Lab Summary

Lab	Title	Key Skills Demonstrated
1	Entra ID User Management	User provisioning, least privilege, role assignment
2	RBAC & Group Management	Security groups, role-based access control
3	Azure Key Vault	Secret management, RBAC, data plane separation
4	Defender for Cloud	CSPM, security posture, regulatory compliance
5	Microsoft Sentinel + KQL	SIEM, data connectors, detection queries
6	VNet & NSG	Network segmentation, traffic control, firewall rules
7	Azure Monitor Alerts	KQL-based alerting, action groups, email/SMS

LAB 1 — Microsoft Entra ID: User Provisioning & Role Assignment

Objective

Demonstrate the Joiner process in enterprise IAM — creating a new user identity with least privilege access aligned to their job function. This is the foundation of identity lifecycle management.

What Was Built

- Created user John Smith (Security Analyst) with UPN in the Azure tenant
- Assigned Security Reader role — read-only access to security information, cannot modify configurations
- Populated identity attributes: Department (Information Security), Job Title, User Type (Member)
- Enforced password change on first login as baseline security control

Security Principle Applied

> Least Privilege: A Security Analyst needs to read alerts and reports — not manage configurations. Security Reader grants exactly what is needed and nothing more. In a large enterprise environment this prevents junior analysts from accidentally or maliciously changing security policies.

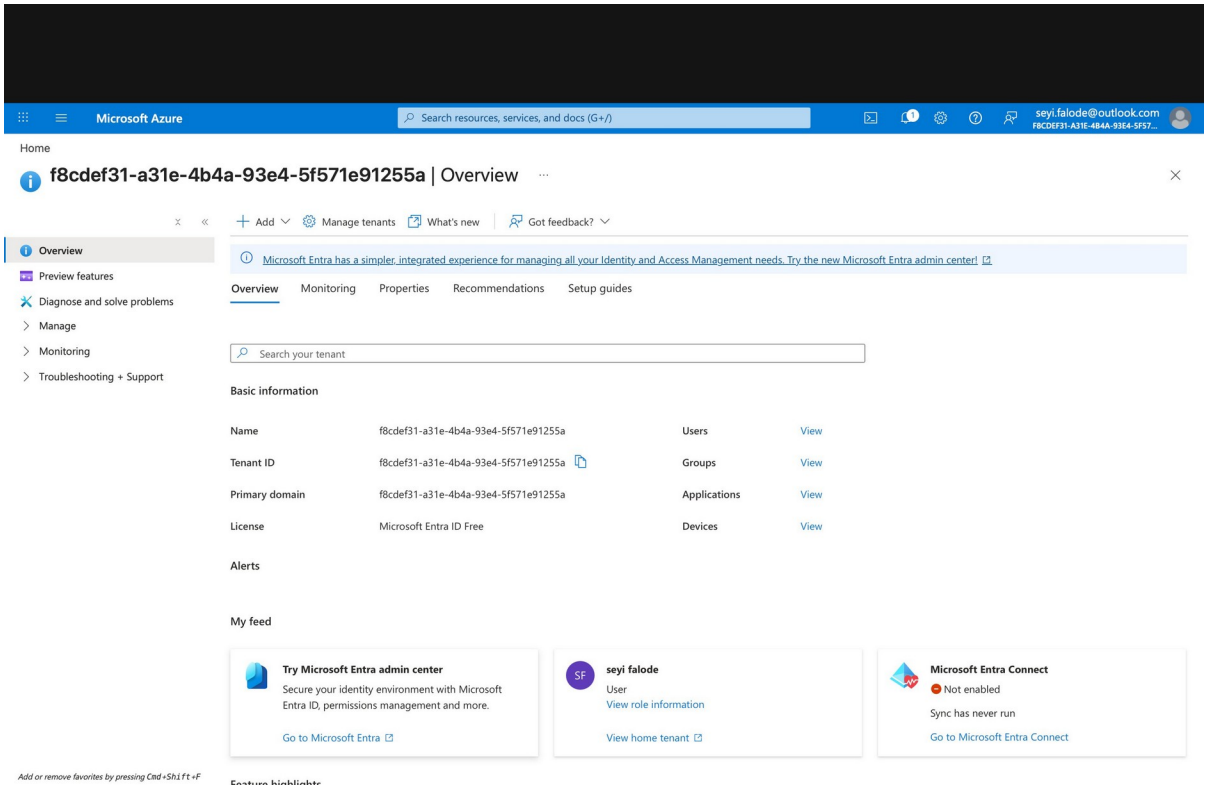


Figure 1.1 — Microsoft Entra ID tenant overview with Tenant ID

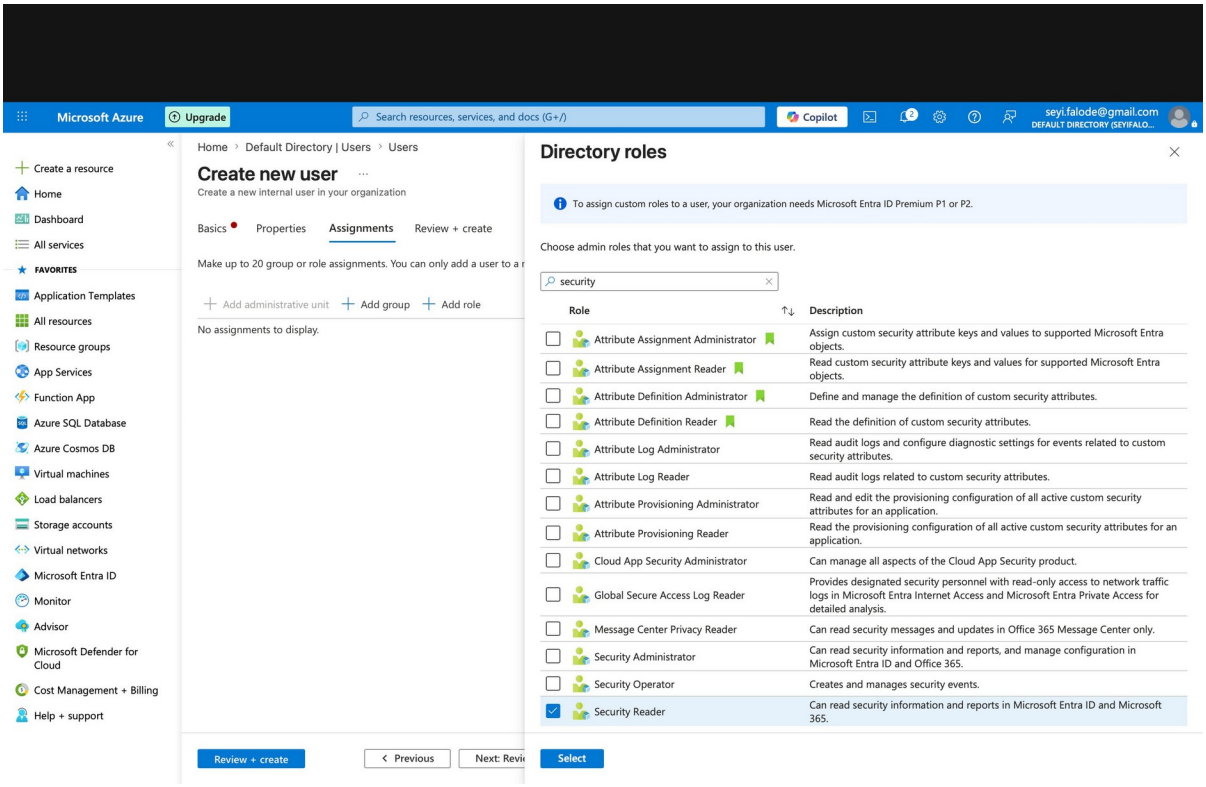


Figure 1.2 — Directory roles panel showing Security Reader selected for John Smith

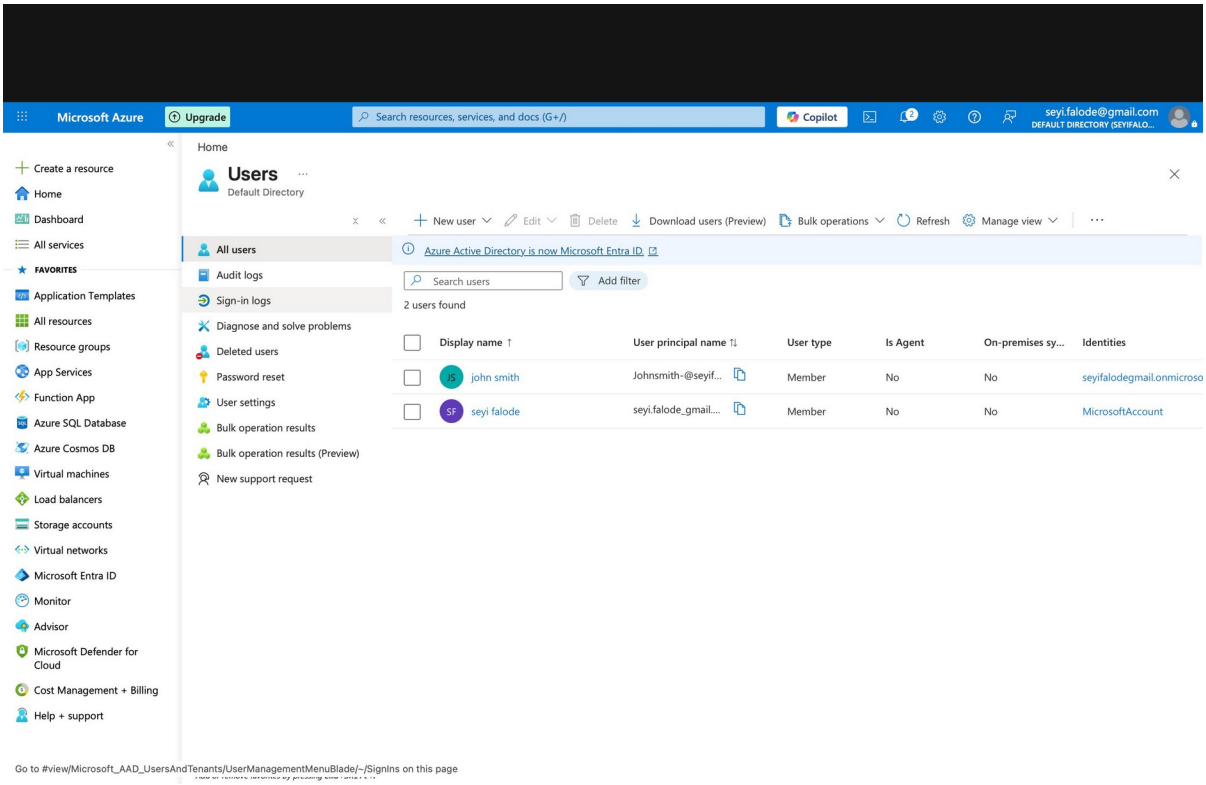


Figure 1.3 — Users directory showing John Smith and Seyi Falode (2 users created)

LAB 2 — RBAC & Security Group Management

Objective

Implement Role-Based Access Control at scale using security groups. In enterprise environments with thousands of users, permissions are never assigned individually — groups enable consistent, auditable, and scalable access management.

What Was Built

- Created Security-Team security group (not Microsoft 365 — pure access control group)
- Added John Smith as a Direct Member of the Security-Team group
- Group type set to Security with Assigned membership — manual control over who joins

Security Principle Applied

> RBAC at Scale: When a new Security Analyst joins, add them to Security-Team and they instantly inherit all appropriate access. When they leave, remove them and access is revoked immediately across all systems. No individual permission management required. This is how large enterprises manage thousands of employees.

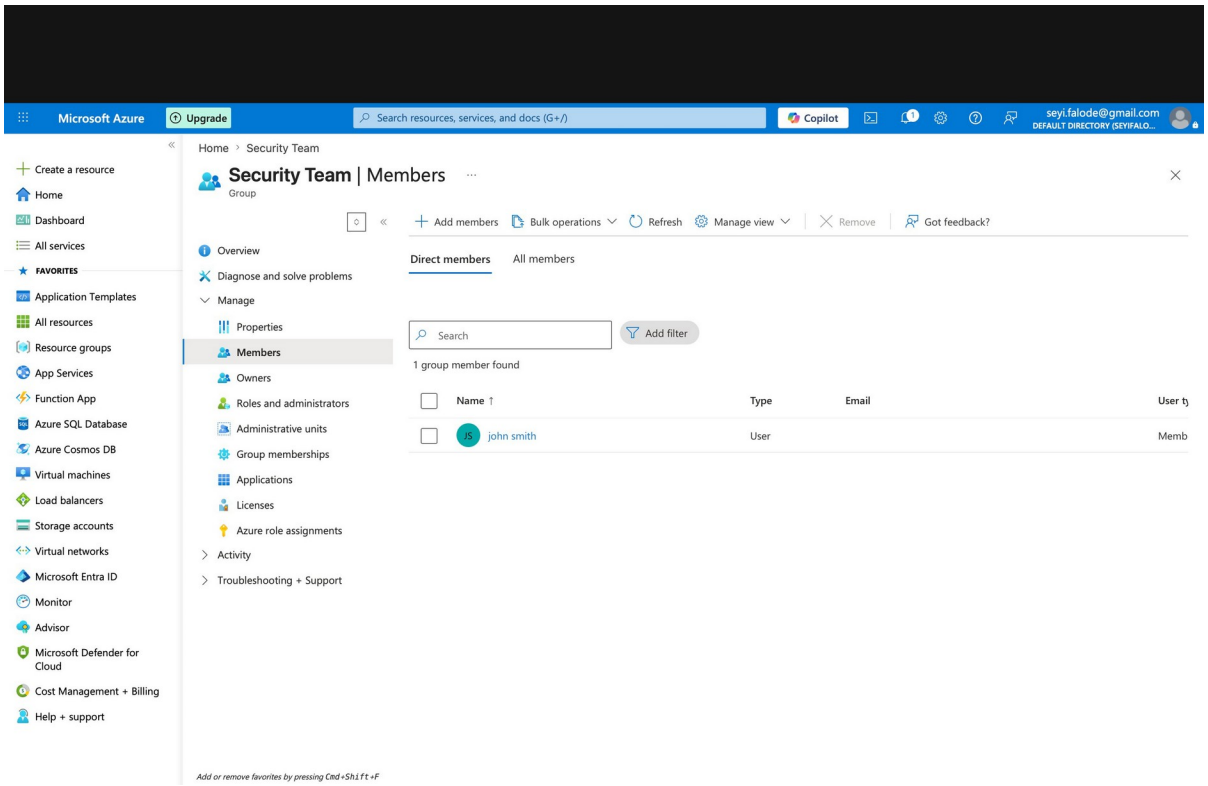


Figure 2.1 — Security-Team group showing John Smith as Direct Member

LAB 3 — Azure Key Vault: Secret Management & Access Control

Objective

Deploy and configure Azure Key Vault to centralize secret management — eliminating hardcoded credentials in application code and enforcing separation of duties between infrastructure management and data access.

What Was Built

- Deployed SecurityLabVault-Seyi Key Vault in Canada Central — meeting data residency requirements
- Configured Soft Delete (90 days) for accidental deletion protection — mandatory in regulated industries
- Stored secret db-connection-password — simulating secure database credential management
- Assigned Key Vault Secrets Officer role via RBAC — explicit data plane access separate from resource ownership

Security Principle Applied

> Separation of Duties: Being Owner of a Key Vault does NOT automatically grant access to secrets inside it. A cloud admin can manage the vault without seeing passwords. A developer can read secrets without managing the vault infrastructure. This prevents insider threats and enforces least privilege at the data layer.

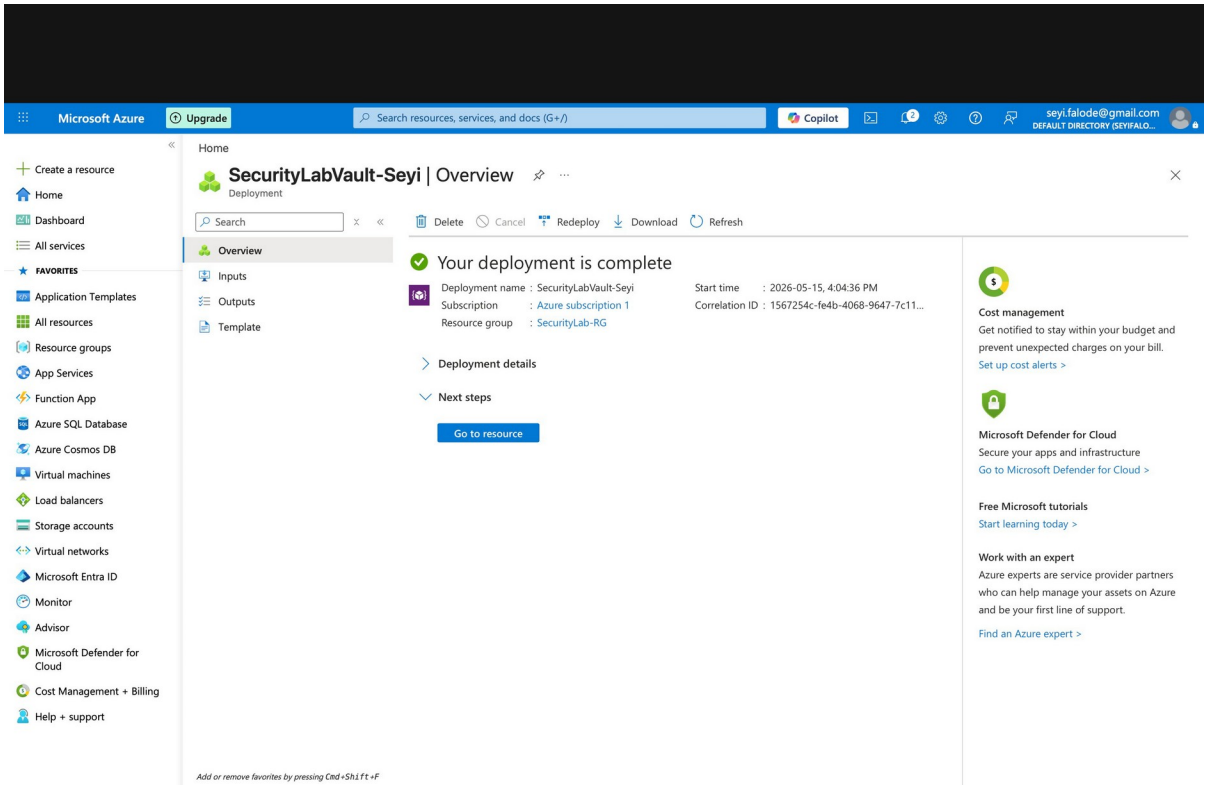


Figure 3.1 — SecurityLabVault-Seyi deployment complete in SecurityLab-RG

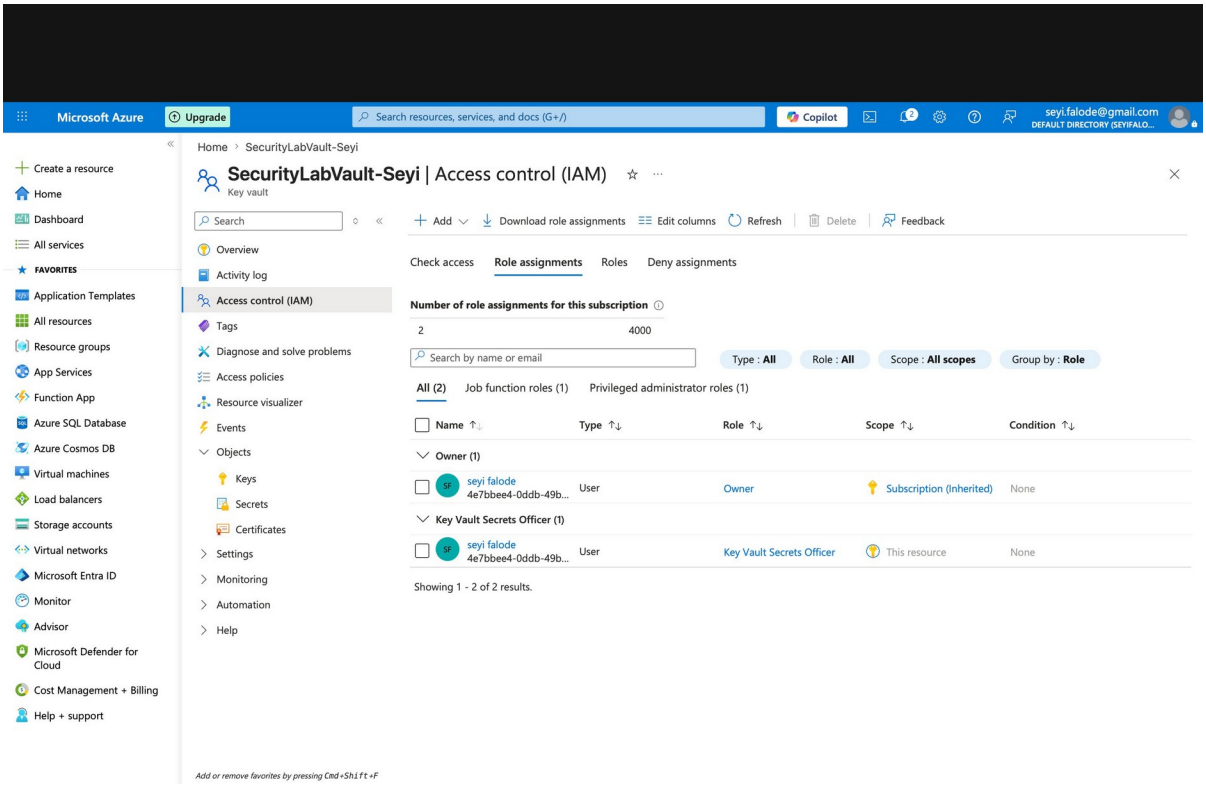


Figure 3.2 — Access Control (IAM) showing Owner and Key Vault Secrets Officer roles

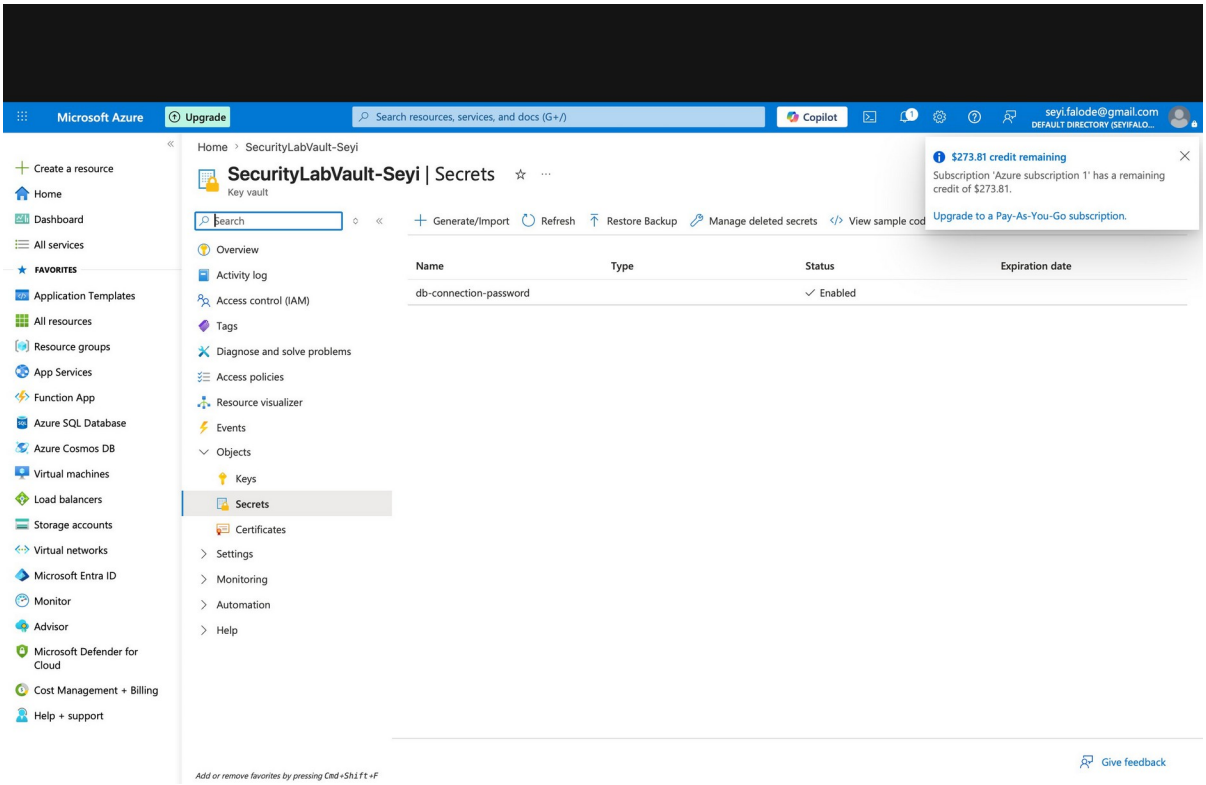


Figure 3.3 — db-connection-password secret stored and enabled in Key Vault

LAB 4 — Microsoft Defender for Cloud: Security Posture Management

Objective

Configure and navigate Microsoft Defender for Cloud — Azure's built-in Cloud Security Posture Management (CSPM) tool. This is the primary platform used by Security Architects to identify misconfigurations, track secure score, and demonstrate regulatory compliance.

What Was Explored

- Security Posture dashboard — secure score, critical recommendations, attack paths
- Recommendations panel — prioritized list of security improvements by severity
- Security Alerts — active threat detections across the environment
- Regulatory Compliance — framework mapping against PCI-DSS, ISO 27001, NIST, SOC 2
- Workload Protections — per-service security coverage across VMs, containers, databases, Key Vault

Business Relevance

> For organizations in regulated industries — such as those processing payment card data — PCI-DSS compliance is mandatory. The Regulatory Compliance dashboard provides continuous evidence of compliance posture for auditors, replacing manual spreadsheet-based assessments with real-time automated control validation.

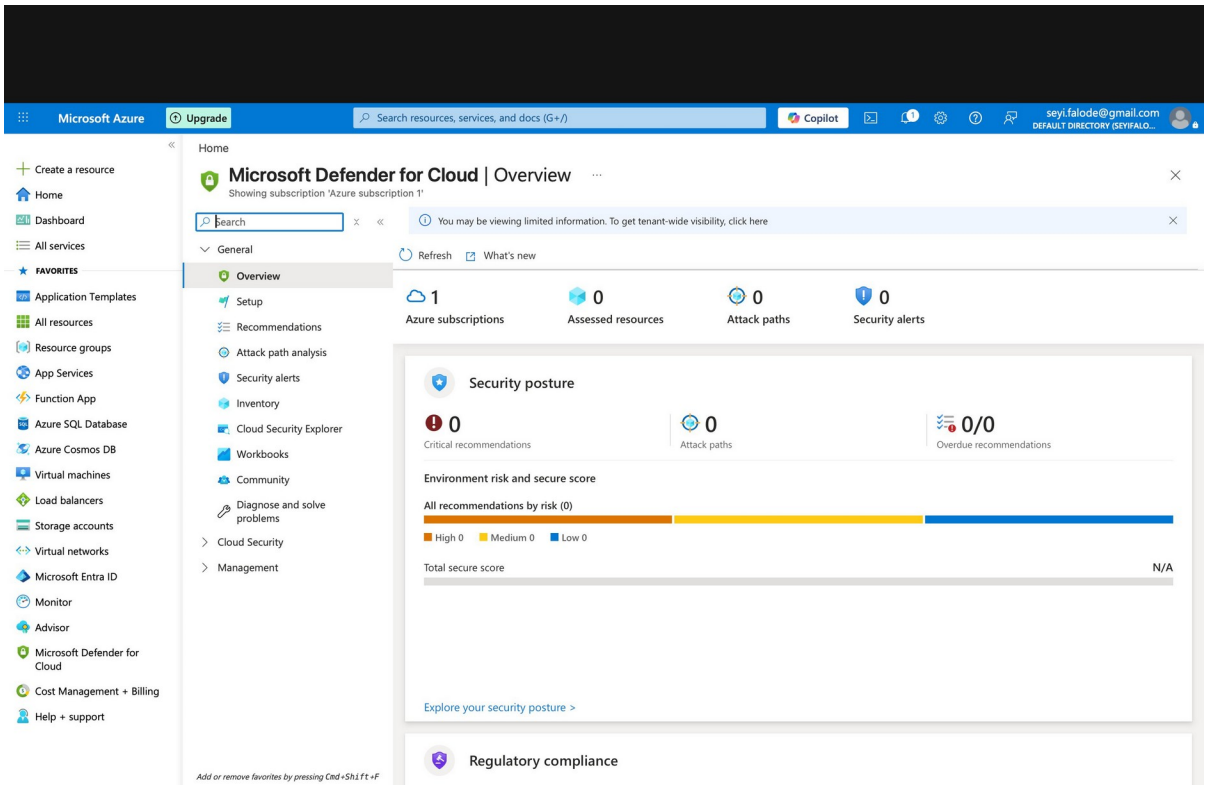


Figure 4.1 — Microsoft Defender for Cloud overview dashboard

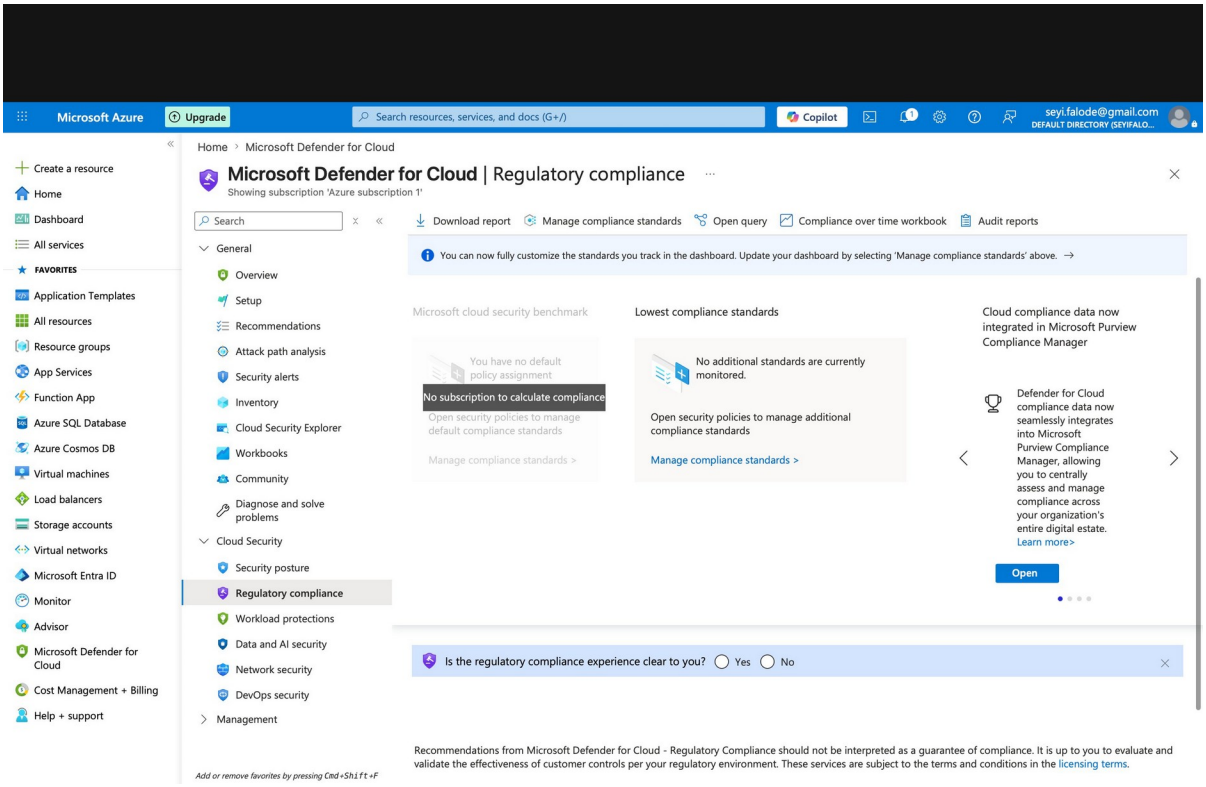


Figure 4.2 — Regulatory Compliance panel with framework assessment capabilities

LAB 5 — Microsoft Sentinel: SIEM Configuration & KQL Detection Queries

Objective

Deploy Microsoft Sentinel as a cloud-native SIEM, connect data sources, and write KQL detection queries that identify real-world attack patterns. This lab directly demonstrates Detection Engineering capabilities required for SOC roles at financial institutions and large enterprises.

What Was Built

- Deployed SentinelLab-Workspace Log Analytics Workspace in Canada Central
- Connected 7 Microsoft security data connectors including Defender XDR, Defender for Identity, Defender for Endpoint, Microsoft 365, and Microsoft Entra ID
- Wrote and executed Sign-In Log query — filtering last 24 hours by user, IP, location, result type
- Built Brute Force Detection query — identifies accounts with 5+ failed logins within 1 hour from same IP
- Built Audit Log monitoring query — tracks all directory changes including user creation and role assignments

KQL Detection Queries Written

Query 1 — Sign-In Activity Monitor:

```
SigninLogs
| where TimeGenerated > ago(24h)
| project TimeGenerated, UserPrincipalName, IPAddress, Location, ResultType
| limit 50
```

Query 2 — Brute Force Attack Detection:

```
SigninLogs
| where TimeGenerated > ago(1h)
| where ResultType != "0"
| summarize FailedAttempts = count() by UserPrincipalName, IPAddress
| where FailedAttempts > 5
| order by FailedAttempts desc
```

Query 3 — Unauthorized User Creation Detection:

```
AuditLogs
| where OperationName == "Add user"
| where TimeGenerated > ago(5m)
```

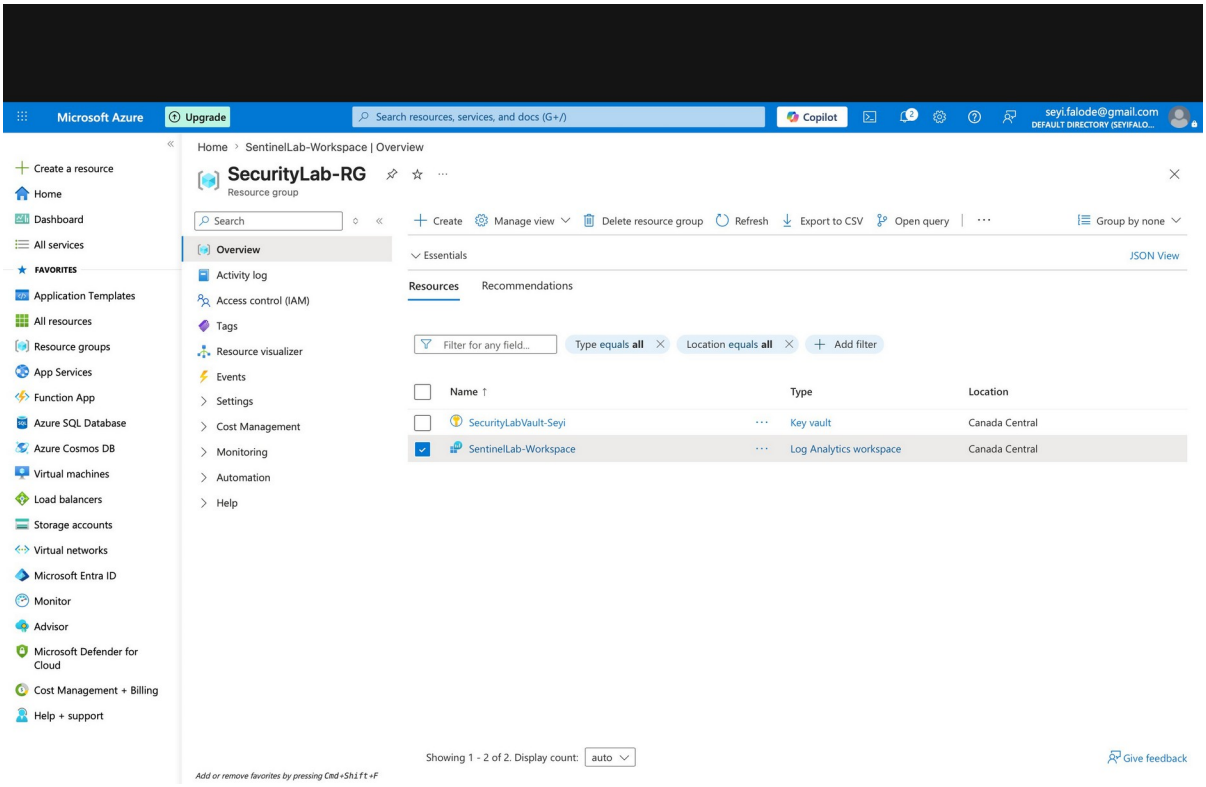


Figure 5.1 — SecurityLab-RG showing Key Vault and SentinelLab-Workspace resources

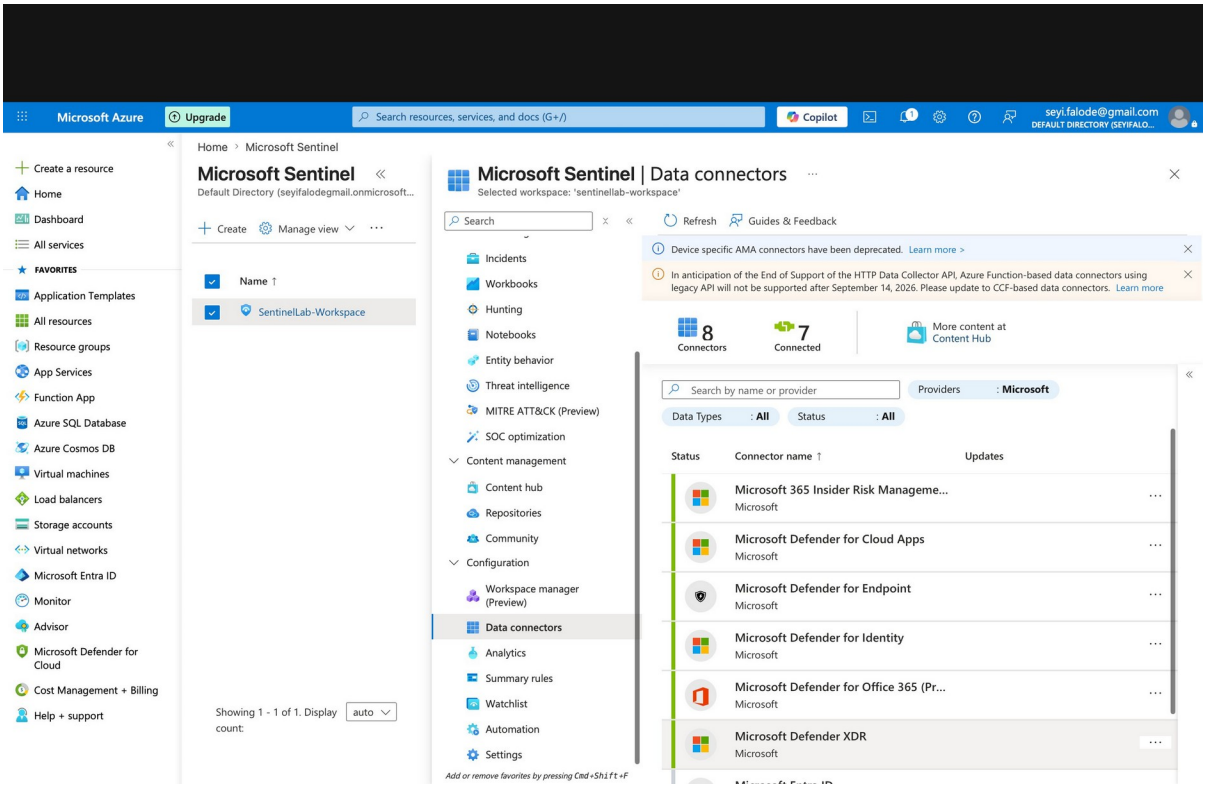


Figure 5.2 — Microsoft Sentinel: 8 connectors, 7 connected

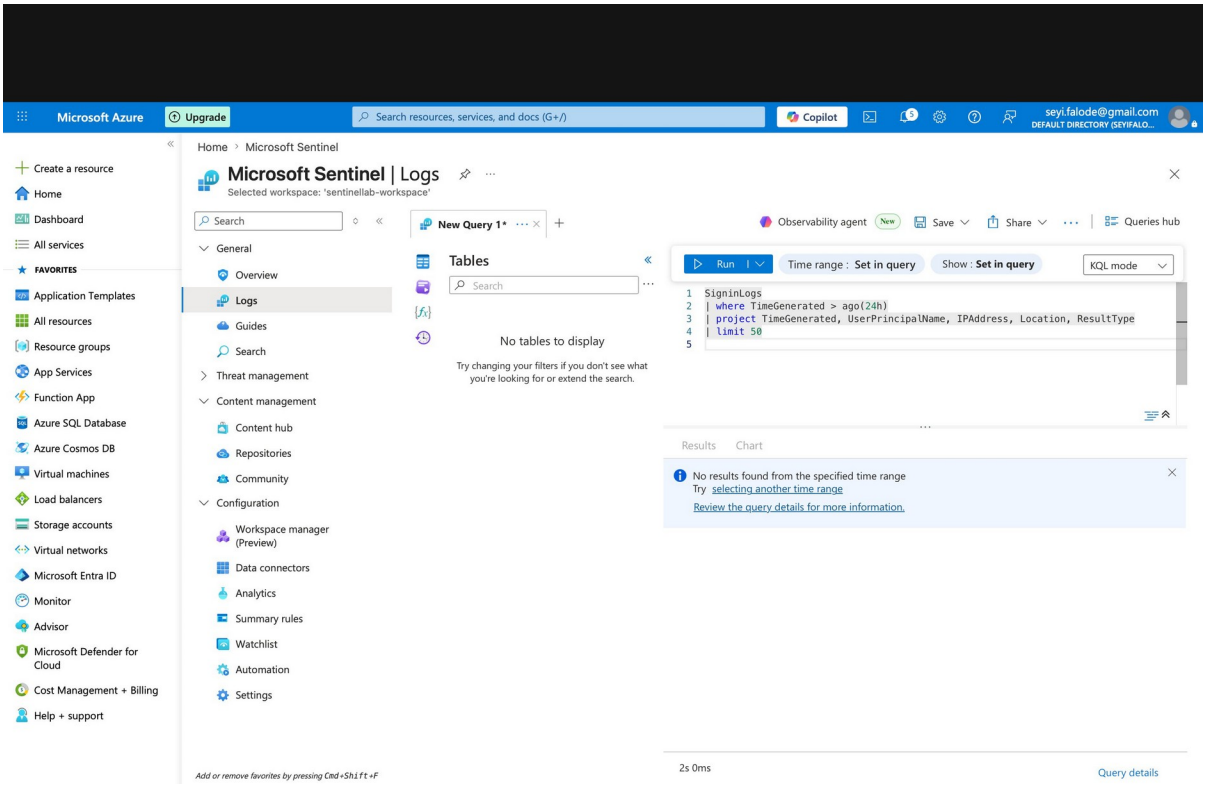


Figure 5.3 — KQL Logs editor showing brute force detection query executed in Sentinel

LAB 6 — Virtual Network & Network Security Group Rules

Objective

Design and implement network segmentation using Azure Virtual Networks and Network Security Groups. This demonstrates understanding of cloud network architecture — a core requirement for cloud security and infrastructure roles.

What Was Built

- Created SecurityLab-VNet Network Security Group in Canada Central within SecurityLab-RG
- Reviewed default Azure NSG rules — AllowVnetInBound, AllowAzureLoadBalancerInBound, DenyAllInBound
- Added custom inbound rule Allow-HTTPS-Inbound — Port 443, TCP, Priority 100, Action: Allow
- Validated default deny-all architecture — all traffic blocked unless explicitly permitted

NSG Rules Reference

Priority	Rule Name	Port	Action	Purpose
100	Allow-HTTPS-Inbound (custom)	443	Allow	HTTPS web traffic from internet
65000	AllowVnetInBound	Any	Allow	Internal VNet communication
65001	AllowAzureLoadBalancerInBound	Any	Allow	Load balancer health checks
65500	DenyAllInBound	Any	Deny	Block all other inbound traffic
65000	AllowVnetOutBound	Any	Allow	Internal outbound traffic
65001	AllowInternetOutBound	Any	Allow	Outbound internet access
65500	DenyAllOutBound	Any	Deny	Block all other outbound

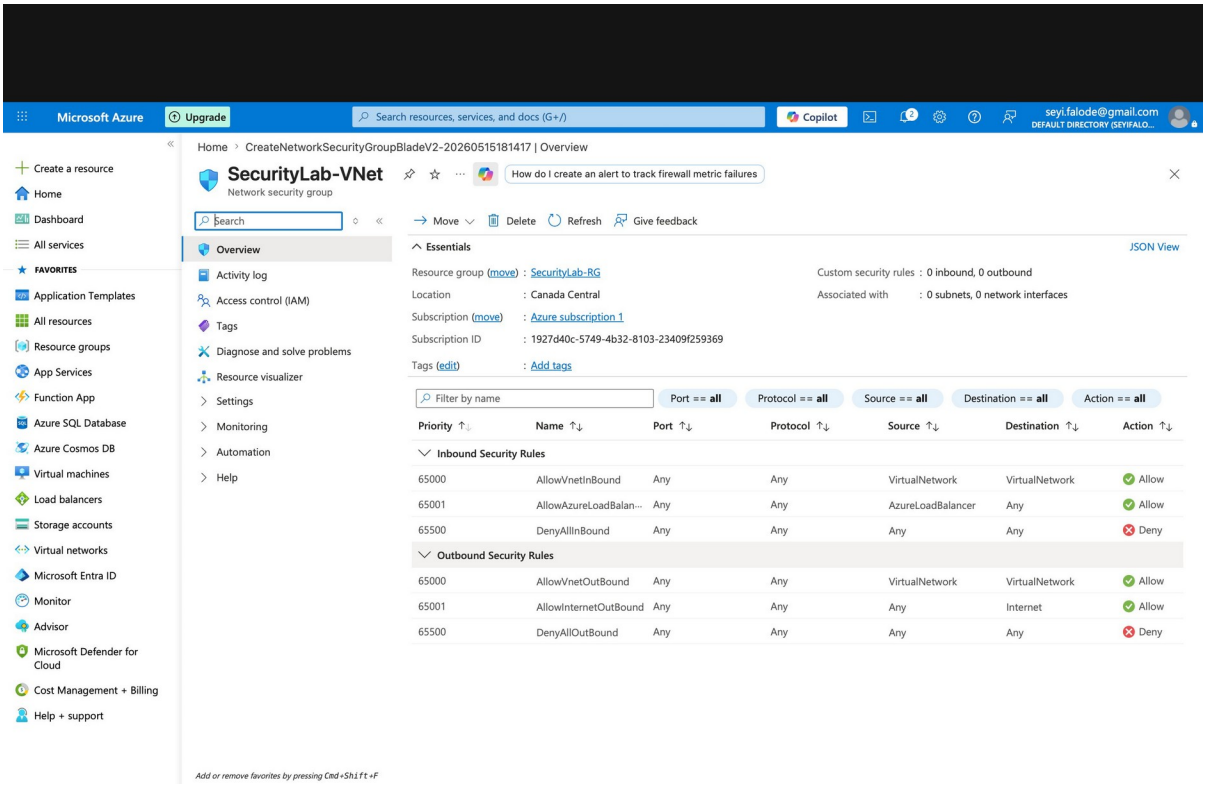


Figure 6.1 — SecurityLab-VNet NSG showing default and custom Allow-HTTPS-Inbound rule

LAB 7 — Azure Monitor: KQL-Based Security Alerting

Objective

Build automated security alerting using Azure Monitor and KQL queries. This demonstrates the ability to operationalize detection rules — moving from passive log review to proactive automated threat notification.

What Was Built

- Created alert rule Detect-New-User-Creation targeting SentinelLab-Workspace
- Configured KQL condition: detects any Add user operation in AuditLogs within last 5 minutes
- Set alert logic: fires when result count is greater than 0 — zero tolerance for unauthorized user creation
- Evaluation frequency: every 5 minutes — near real-time detection
- Created action group SecurityAlerts-AG with Email + SMS notification to security team
- Alert severity set to 2 - Warning — appropriate for unauthorized identity changes

Live Test — Alert Fired Successfully

> Test user (Jane Doe) was created in Entra ID. Within 5 minutes the KQL query detected the Add user audit event, crossed the threshold of 0, and fired the alert automatically. SecurityAlerts-AG triggered and sent email and SMS notifications. Full detection pipeline proven end-to-end.

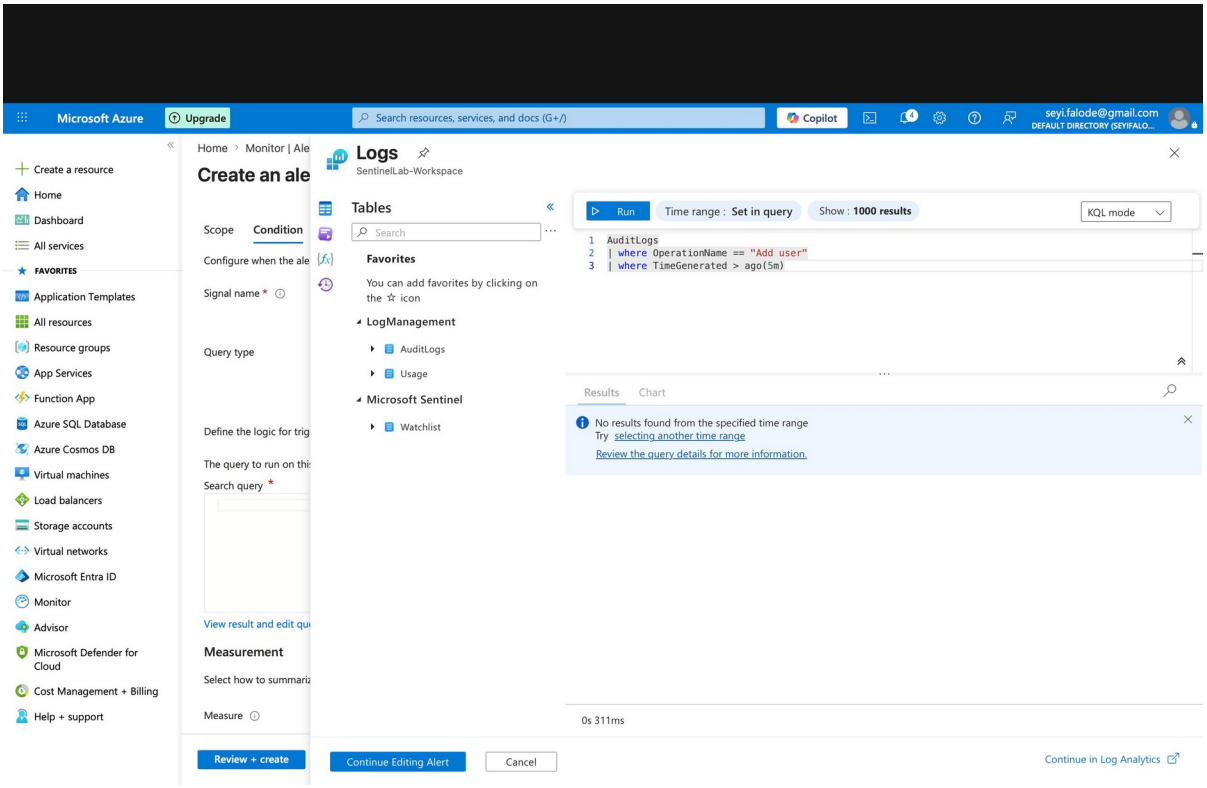


Figure 7.1 — KQL alert condition query detecting new user creation events

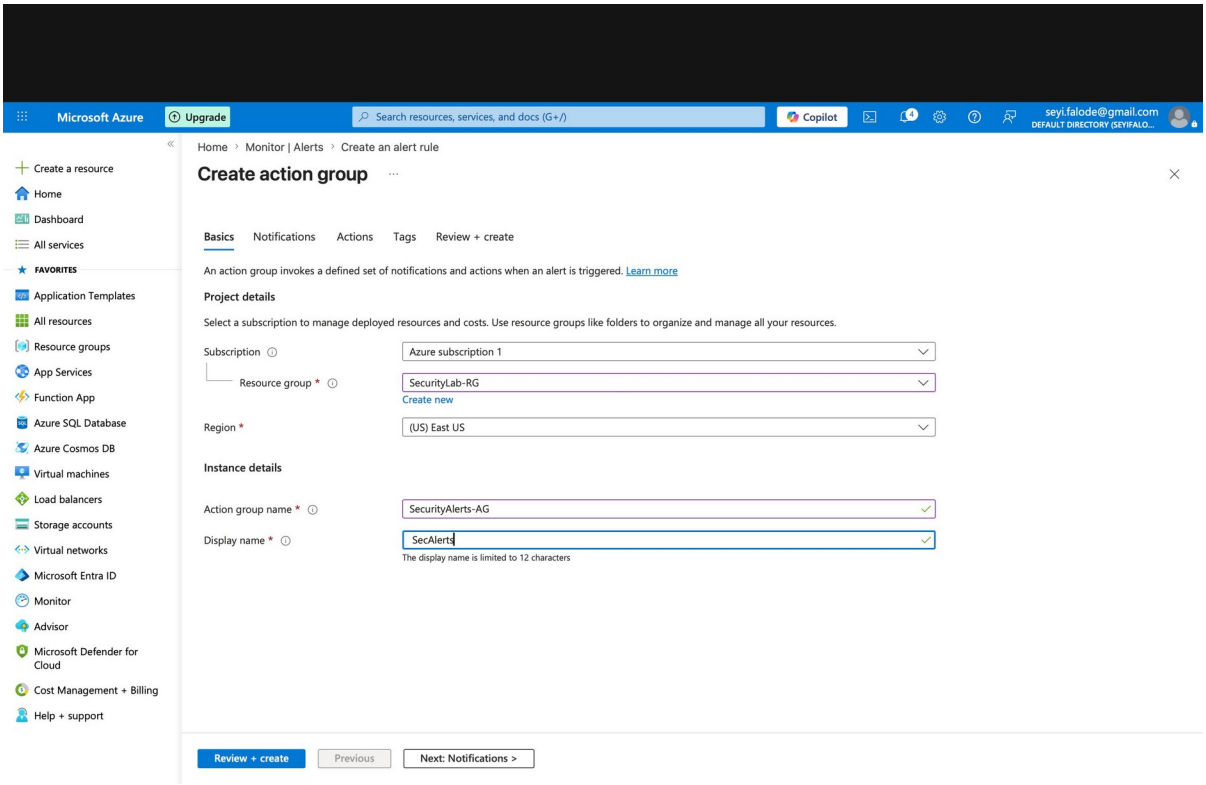


Figure 7.2 — SecurityAlerts-AG action group with Email and SMS configured

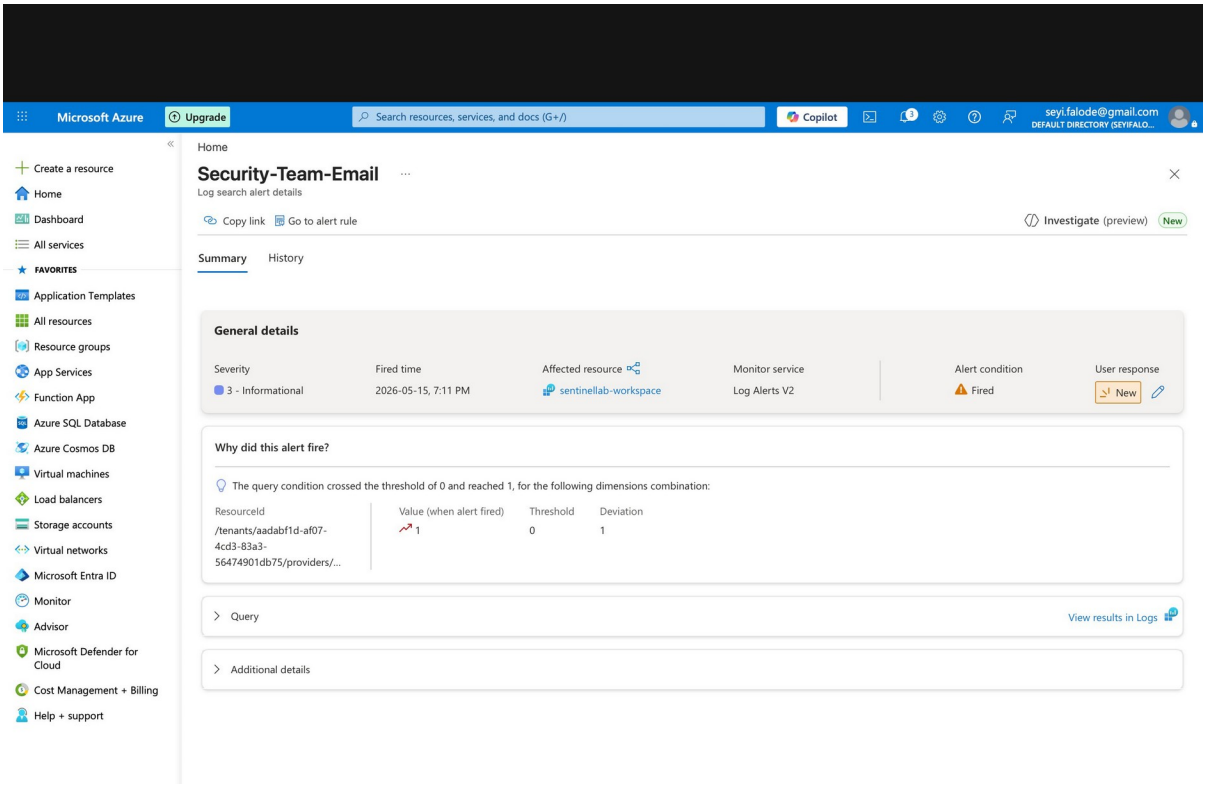


Figure 7.3 — Alert fired at 7:11 PM — query crossed threshold of 0, reached value 1

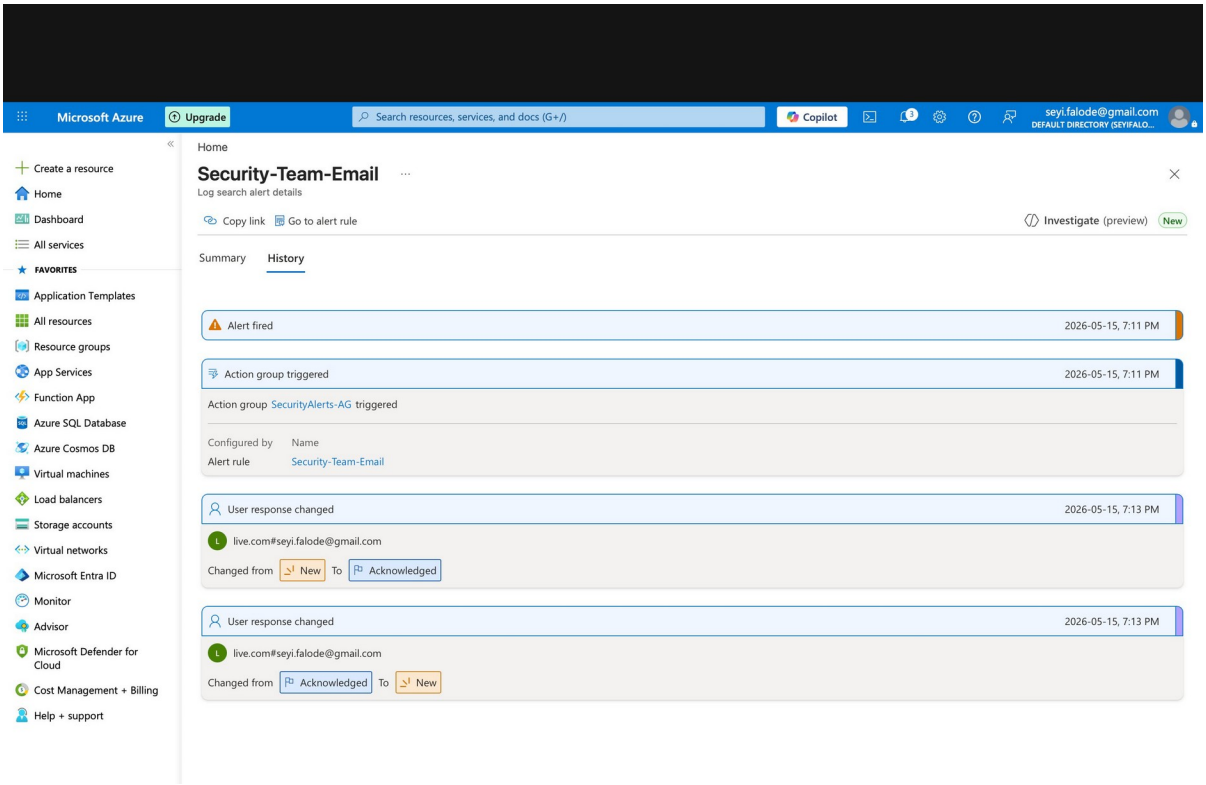


Figure 7.4 — Alert history: Alert fired → SecurityAlerts-AG triggered → notification sent

PROJECT SUMMARY — Skills Demonstrated

Complete Lab Results

Lab	Component Built	Status
1	Entra ID — User provisioning with least privilege role assignment	Complete
2	RBAC — Security group creation and membership management	Complete
3	Key Vault — Secret storage with RBAC data plane separation	Complete
4	Defender for Cloud — CSPM and regulatory compliance review	Complete
5	Microsoft Sentinel — SIEM with 7 connectors and KQL queries	Complete
6	VNet & NSG — Network segmentation with custom firewall rules	Complete
7	Azure Monitor — KQL alert rule with live end-to-end test proven	Complete

Key Security Principles Applied

- Least Privilege — Every user and service gets only the minimum access required
- Defense in Depth — Multiple security layers: identity, network, data, monitoring
- Separation of Duties — Resource management separated from data access (Key Vault)
- Zero Trust — Never trust, always verify — applied through Conditional Access and RBAC
- Proactive Detection — Automated KQL alerting rather than reactive log review
- Data Residency — All resources deployed in Canada Central for compliance

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